🚆 PASHUPATASTRA — TEAM PRESENTATION

AI-Assisted Railway Maintenance Scheduling & Disruption Recovery

Workstream: Railway Domain Modeling & Synthetic Data

Presenter / Owner: Darshini (`feature/domain-data`)

Deliverable: Phase 1 Domain Review & Master Railway Datasets

Target: Smart India Hackathon 2026 — Domain Freeze

1. Domain & Data	2. ML Risk Scorer	3. CP-SAT Solver	4. FastAPI Backend	5. Operations UI	6. Disruption Sim
Darshini (You)	Ayush	Tyagi	Aryan	Archit	Tirth

1. Domain Decisions for Team Freeze (Summary for Aryan & Tyagi)

> **Key Rule:** Zero breaking changes for the CP-SAT solver. `track_id: str` is preserved while rich Indian Railways metadata is structured in the data layer.

Domain Dimension	Agreed Phase 1 Standard	Impact on Teammates
Track Representation	String `track_id` (`"UP-1"`, `"DOWN-1"`) + Geographic metadata dictionary (corridor, direction, km start/end, speed limit).	**Tyagi (Solver):** No constraint changes. **Archit (UI):** Groups tracks by UP/DOWN.
Work Categories	6 Indian Railways maintenance categories (Renewals, Tamping, OHE, Signalling, Inspection, Emergency).	**Ayush (Scorer):** Computes distinct risk profiles. **Tyagi:** Realistic durations.
Machinery & Resources	Model shared heavy machines via `mutual_exclusion_group` (e.g. `BCM_01`, `CSM_01`) rather than crew rostering.	Prevents CP-SAT solver timeouts and artificial infeasibility during live demo.
Train Timetable	Train traffic modeled as **Discrete Possession Windows** per track (Night, Midday, Evening slots).	Solver enforces: `start >= window.start` and `end <= window.end`.
Disruption Scenarios	4 deterministic disruption triggers (Track Unavailable, Emergency Injection, Possession Shortened, Machine Breakdown).	**Tirth (Simulation):** Drives repeatable PLAN → DISRUPT → RECOVER demo flow.

2. Indian Railways Corridor Master Data

Track Layout (New Delhi – Ghaziabad Section, 40 KM)

Track ID	Direction	Line Type	Speed Limit	Traction	Daily Density	Headway Buffer
`UP-1`	UP (To Delhi)	Up Main Line (Passenger & Express)	130 km/h	25kV AC	85 trains/day	15 min
`DOWN-1`	DOWN (From Delhi)	Down Main Line (Passenger & Express)	130 km/h	25kV AC	90 trains/day	15 min
`UP-2`	UP (To Delhi)	Up Loop / Freight Line	100 km/h	25kV AC	45 trains/day	15 min
`DOWN-2`	DOWN (From Delhi)	Down Loop / Freight Line	100 km/h	25kV AC	50 trains/day	15 min

Operational Possession Windows (24-Hour Horizon = 1440 Minutes)

Window ID	Track	Window Type	Clock Time	Minute Timeline	Allocated Maintenance Work
`POS-UP1-NIGHT`	`UP-1`	NIGHT_TRAFFIC_BLOCK	00:30 – 05:00	**0030 – 0300** (270 min)	Heavy Track Renewal (CTR) & Ballast Tamping
`POS-UP1-MIDDAY`	`UP-1`	MIDDAY_MAINTENANCE	11:30 – 14:30	**0690 – 0870** (180 min)	USFD Inspection & OHE Isolations
`POS-UP1-EVENING`	`UP-1`	EVENING_OFF_PEAK	21:00 – 23:30	**1260 – 1410** (150 min)	Point Machines & Signalling Interlocking
`POS-DN1-NIGHT`	`DOWN-1`	NIGHT_TRAFFIC_BLOCK	00:30 – 05:00	**0030 – 0300** (270 min)	Heavy Track Renewal (CTR) & Ballast Tamping
`POS-DN1-MIDDAY`	`DOWN-1`	MIDDAY_MAINTENANCE	11:30 – 14:30	**0690 – 0870** (180 min)	USFD Inspection & OHE Isolations
`POS-DN1-EVENING`	`DOWN-1`	EVENING_OFF_PEAK	21:00 – 23:30	**1260 – 1410** (150 min)	Point Machines & Signalling Interlocking

3. Scenario A: Baseline Mainline Dataset (`corridor_a_blocks.json`)

Corridor: `CORRIDOR_A` | **Tracks:** `UP-1`, `DOWN-1` | **Candidates:** 12 Blocks | **Clean Feasible Solution**

Block ID	Track	Asset ID	Work Category	Dur	Pri	Risk	Time Window (Clock)	Machine Group	Precedence
BLK-001	DOWN-1	AST-DN-OHE-013	OHE_MAINTENANCE	120 min	0.79	0.63	11:00 – 15:30 (0660-0930)	TOWER_WAGON_01	—
BLK-002	DOWN-1	AST-DN-RAI-015	BALLAST_TAMPING	105 min	0.50	0.50	00:30 – 06:00 (0030-0360)	CSM_TAMPER_01	—
BLK-003	DOWN-1	AST-DN-RAI-014	TRACK_RENEWAL	210 min	0.59	0.66	00:30 – 06:00 (0030-0360)	BCM_MACHINE_01	—
BLK-004	DOWN-1	AST-DN-RAI-009	BALLAST_TAMPING	150 min	0.59	0.59	04:00 – 07:30 (0240-0450)	CSM_TAMPER_01	BLK-003 🔗
BLK-005	UP-1	AST-UP-OHE-004	OHE_MAINTENANCE	75 min	0.56	0.52	00:30 – 23:30 (0030-1410)	TOWER_WAGON_01	—
BLK-006	DOWN-1	AST-DN-OHE-016	OHE_MAINTENANCE	120 min	0.54	0.49	00:30 – 23:30 (0030-1410)	TOWER_WAGON_01	—
BLK-007	DOWN-1	AST-DN-OHE-011	OHE_MAINTENANCE	120 min	0.79	0.53	11:00 – 15:30 (0660-0930)	TOWER_WAGON_01	—
BLK-008	UP-1	AST-UP-RAI-001	TRACK_RENEWAL	210 min	0.78	0.77	00:30 – 06:00 (0030-0360)	BCM_MACHINE_01	—
BLK-009	UP-1	AST-UP-TUR-007	SIGNALLING_INTERLOCK	105 min	0.42	0.44	00:30 – 23:30 (0030-1410)	S_T_GANG_01	—
BLK-010	UP-1	AST-UP-RAI-008	ROUTINE_INSPECTION	45 min	0.42	0.46	11:00 – 15:30 (0660-0930)	—	—
BLK-011	UP-1	AST-UP-SIG-002	SIGNALLING_INTERLOCK	75 min	0.65	0.59	00:30 – 23:30 (0030-1410)	S_T_GANG_01	—
BLK-012	UP-1	AST-UP-TUR-006	SIGNALLING_INTERLOCK	75 min	0.57	0.50	00:30 – 23:30 (0030-1410)	S_T_GANG_01	—

4. Scenario B & Scenario C Overview

Scenario B: Dense Quadruple Corridor (`corridor_b_dense.json`)

Corridor: `CORRIDOR_B_DENSE` | **Tracks:** `UP-1`, `UP-2`, `DOWN-1`, `DOWN-2` | **Candidates:** 24 Blocks
Purpose: Simulates high-density trunk routes (New Delhi – Kanpur). 4 heavy machine groups competing across 4 tracks with strict headway buffers, testing CP-SAT objective trade-offs and capacity limits.

Scenario C: Disruption & Recovery Testbed (`corridor_c_disrupted.json`)

Corridor: `CORRIDOR_C_DISRUPTED` | **Tracks:** `UP-1`, `DOWN-1` | **Candidates:** 14 Blocks
Pre-Committed Blocks (Locked):

- `BLK-001`: `DOWN-1`, 01:00 – 03:00 (0060–0180 min), **COMMITTED** (Cannot be moved).
- `BLK-002`: `DOWN-1`, 01:30 – 03:15 (0090–0195 min), **COMMITTED** (Cannot be moved).

Disruption Event: `TRACK_UNAVAILABLE` on `UP-1` from 02:00 to 05:00. Tirth's simulation will verify that Tyagi's optimizer re-routes unaffected blocks while protecting `BLK-001` & `BLK-002`.

5. Automated Test Results & Deliverable Locations

> **Test Execution Result:** `10/10 Unit Tests Passed in 0.049s` (Seed reproducibility, schema validity, duration validity, dependency integrity, fixture loading).

Deliverable	File Path	Primary Downstream Consumer
Domain Review Doc	`DOMAIN_REVIEW.md`	Whole Team (Aryan, Tyagi, Ayush, Tirth, Archit)
Domain Models	`backend/app/data/models.py`	Ayush (ML Scorer) & Aryan (API)
Synthetic Generator	`backend/app/data/generator.py`	Tyagi (Optimizer) & Tirth (Simulation)
JSON Fixtures	`backend/app/data/fixtures/*.json`	Archit (Frontend Dashboard) & Tyagi
Unit Tests	`backend/tests/test_generator.py`	Continuous Integration / CI Validation